

Sheet 7

outstanding

1. cph: 200, 100.

WTP: £10, £4, £2
cph

A B C

a) Bidding truthfully, bids are (£10, £4, £2).

A and B win slots and pay ~~10~~ 42 per click. C pays 0.

So $\Pi_C = 0$, $\Pi_A = 200 \times (10 - 4) = 1200$;

$\Pi_B = 100 \times (4 - 2) = 200$ per hour.

b) We need each company to be BRing to the others! no strict incentive to deviate, holding others fixed

C: - if they bid $c > 4$ they'd win, but make a net loss, ^{-2 per click}
- if some $c \leq 4$ but $\neq 2$, profits remain unchanged at 0.

B: - if they bid $b < 2$, they lose auction and get 0 profit.
- if they bid $b > 10$, they win 1st place but make a net loss of $4 - 10 = -6$ per click
- if they bid $2 \leq b \leq 10$, $b \neq 4$ no change in profits.

A: - if $a < 2$ win ~~not~~ lose auction for 0 profit.
- $2 \leq a < 4$, win 2nd place with net profit per click of £8, but only £800 ph, so worse.
- $a \geq 4$, no change

So no firm has strict unilateral incentive to deviate $\Rightarrow$ truth-telling PSNE.

c) Suppose 2nd click-rate is £175ph, then A's net profit from $2 \leq a < 4$ would be $(10 - 2) \times 175 = 1400 > 1200$, So not BR to bid $a = 10$, hence truth-telling not NE.

d) Like a standard VCG auction, agents here pay the bid of whoever they priced out. But the payment is per click, where each "good" has different click rates.

Under normal VCG

Good!

$$(200-100) \times 4 + (100-0) \times 2$$

✓ A's payment per hour should be ~~(100-0) \times 2~~ ~~(200-0) \times 2~~ = 600 as the ^{difference in gross} surplus B and C would've got had A not participated. But A actually pays 800ph, which is too much. Implicitly it's assuming A fully displaces the next-highest bidder to get 0 cph, when in reality B loses only 100.

2.a) A matching μ is student-optimal and stable if it's stable := [non-wasteful, free of justified envy, individually rational], and weakly preferred by all students to any other stable matching

DA delivers such a matching.

$$\begin{array}{c|c} S_1 & i_3 \\ S_2 & i_5 \\ S_3 & i_1 \end{array} \quad \begin{array}{c} i_2 \\ i_4 \end{array}$$

$$\begin{array}{c|c} S_1 & i_3 \quad i_2 \\ S_2 & i_5 \\ S_3 & i_1 \quad i_4 \end{array}$$

$$\begin{array}{c|c} S_1 & i_2 \\ S_2 & i_5 \quad i_3 \\ S_3 & i_1 \quad i_4 \end{array}$$

✓ The students who didn't get their first preferences are i_2 and i_3 only. So you couldn't change any other student's school without making someone worse off. And i_2 and i_3 got their second prefs, so you couldn't swap them ^{with each other} either. S_0 in this case (but not in general) it's efficient.

b) First step is as above. But i_2 could not then successfully propose to 3, as it's full, so they end up at S_2 .

$$S_1: i_3, \quad S_2: i_5, i_2, \quad S_3: i_1, i_4.$$

✓ ~~Therefore~~ This isn't stable, since ~~there's~~ there's justified envy: i_2 strictly prefers S_1 to S_2 and they're higher priority there than i_3 , who was admitted instead.

2)

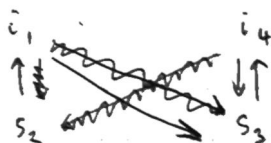
c)



$s_1: i_3$

$s_2: i_2$

$s_3: i_4$



$s_3: i_4$

and then i_1 left with s_2 .

So $s_1: i_3$, $s_2: i_1, i_2$, $s_3: i_2, i_4$

Only i_1 and i_2 didn't get their first preference. i_1 has justified envy of i_2 , so not stable.

3 a)

		α	$1-\alpha$
		H	D
$1-p$	H	-1, -1	2, 0
D	D	0, 2	1, 1

Two ISNE by BR payoffs:
(D, H) and (H, D).

Also one MSNE. Let row, col play H

w.p. p, q . Then by indifference principle:

$$(-1) \cdot q + 2(1-q) = 1-q \Rightarrow p = q = \frac{1}{2}$$

$$\text{and } -1 \cdot p + 2(1-p) = 1-p \Rightarrow$$

so $(\frac{1}{2} D + \frac{1}{2} H, \frac{1}{2} D + \frac{1}{2} H)$ also NE.

None others as strict BRs so can't have i mix, -i pure.

- An ESS is a symmetric NE α^* s.t. α^* is strictly better against $\alpha^* \neq \beta = BR(\alpha^*)$ than β is against itself.
- Since every 2x2 symmetric game has an ESS, H-D must, and it has to be the MSNE identified, since BNE not symmetric.

Note that any mix of H and D is a BR to α^* , given it's a mixed strategy. So consider $\beta := p H + (1-p) D$.

$$v(\beta, \beta) = -p^2 + p(1-p) \cdot 2 + (1-p)^2 = -2p^2 + 1$$

$$v(\alpha^*, \beta) = -\frac{1}{2}p + \frac{1}{2}(1-p) \cdot 2 + \frac{1}{2}(1-p) = \frac{3}{4} - 2p$$

so α^* is strictly better

than β is against itself

a) Consider now the difference, $\frac{3}{2} - 2p + 2p^2 - 1$.

$$= 2(p - \frac{1}{2})^2 - \frac{1}{2} + \frac{3}{2} - 1 = 2(p - \frac{1}{2})^2 > 0 \text{ for } p \neq \frac{1}{2}$$

So indeed α^* is ESS.

b) The replicator dynamic is defined as

$$\frac{d p_i(t)}{d t} = p_i(t) \cdot [\underbrace{\pi_i(t)}_{\text{payoff from strat. } i} - \underbrace{\Pi(t)}_{\text{population avg payoff}}]$$

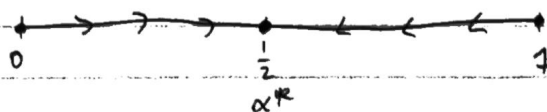
payoff from strat. i

population avg payoff

letting $\lambda = 2 \ln 2$. i.e., relative growth of proportion following $i \propto$ how much better i is than average strategy.

	H (p)	D ($1-p$)	π_i for $i \in \{H, D\}$
H	-1	2	$-1 \cdot p + 2 \cdot (1-p) = 2 - 3p$
D	0	1	$0 \cdot p + 1 \cdot (1-p) = 1-p$

Note that p increases iff $2 - 3p > 1 - p \Leftrightarrow p < \frac{1}{2}$.



$p = \frac{1}{2}$, i.e. α^* the ESS, is the globally stable rest point. Provided the starting population is non-degenerate, we'll end up with half the population playing each of H and D.

[Is it by assumption that agents only play pure strategies? Slider says every ESS is locally stable rest point, but they seem like quite different objects — ESS is a mixed strat, rest point is a proportion of population playing pure strat.]

Correct. They're very different objects.

Maybe better to say: the population state corresponding to α^* is our globally stable rest point, and both have the same outcome: half of all actions are H, half are D.

Yes

		^{1-q}	^{1-q}
	^{1-p}	H	D
P	H	-1, -1	2, 0
1-p	D	0, 2	1, 1

c) The $\mathbb{E}$ payoff from row playing H against col is

$$-1 \cdot q + (1-q) \cdot 2 = 2 - 3q$$

and playing D is $0 \cdot 2 + 1 \cdot (1-q) = 1-q$

By LIE, avg payoff from row against col is

$$p(2-3q) + (1-p)(1-q)$$

So we need

$$\dot{p}/p = (2-3q) - p(2-3q) - (1-p)(1-q)$$

$$= (1-p)(2-3q) - (1-p)(1-q)$$

$$= (1-p)(2-3q+q-1)$$

$$= (1-p)(1-2q) \text{ as required, and symmetric for } \dot{q}/q.$$

So we have a pair of coupled ODEs.

(i) if $p > 1/2$ and $q < 1/2$ then $1-2p < 0$, $1-2q > 0$. So p would shrink and q would grow, and this continues to PSNE $(0, 1)$.

(ii) Rates of chg equal in both, so $\dot{p}/p = (1-p)(1-2p)$. If $p < 1/2$ then grows to rest point at $1/2$, else shrinks, then. So converge at MSNE.

(iii) Both grow; but q will grow faster because

Consider e.g. $q = 0.3$, $p = 0.4$.

Then $\dot{q} = 0.042$, $\dot{p} = 0.046$. So p grows faster and we'll end up in case (i).

In the long term we'd expect to arrive at one of the PSNE by (i) since (ii) is a knife-edge result - not locally stable.

Great!

4a)

		q	$1-q$
		L	R
P	L	4, 4	0, 1
1-P	R	1, 0	1, 1

Two PSNE: (L, L) and (R, R).
One MSNE (both mixing):

$$4q + 0 \cdot (1-q) = 1 \cdot q + 1 \cdot (1-q)$$

$\Rightarrow q = 1/4$ and must be symmetric so

MSNE: $(1/4 L + 3/4 R, 1/4 L + 3/4 R)$ also NE.

Clearly (L, L) and (R, R) are ESS because they're strict BR to themselves, so form strict PSNE and mutant couldn't invade.

The MSNE isn't ESS because it can be invaded by L:

- L is a BR against the mixed strategy, by indifference principle

- But its expected payoff is $4 \cdot 1/4 + 0 \cdot 3/4 = 1$, whereas

MSNE against itself expected payoff of MSNE against L is $4 \cdot 1/4 + 1 \cdot 3/4 = 7/4 < 4$, which is $u(L, L)$.

b)

		p	$1-p$	
		L	R	π_i
K	L	4	0	$4p$
R	R	1	1	1

$\dot{p}/p = \pi_L(t) - \pi_R(t) > 0$
 $\Leftrightarrow \pi_L > \pi_R$

i.e. p increases iff $p > 1/4$.



The ~~MSNE~~ ~~(1/4, 1/4)~~ has a larger basin of attraction, but if you start with $p < 1/4$ you converge to R. $p = 1/4$ is a rest point but not locally stable, corresponds to MSNE (not ESS).

c) In the long run, we'd expect the system to spend an increasingly large fraction of its time at L (i.e. all agents play L) as $\epsilon \rightarrow 0$. This is due to stochastic stability: the process will almost always be in the state which has the largest basin of attraction with $\epsilon = 0$. Here, the two absorbing states are R

and L , and by b) clearly L has the larger basin,
(and $1/4$) $(1/4, 1]$. ✓

This corresponds to the risk-dominant NE of the coordination game - that is, the eq^{*} made up of the strategies that're BR to an opponent mixing 50:50, which is just (L, L) .

Note that from R , we need about $1/4$ mistakes to reach L 's basin of attraction, but need $3/4$ from L to R 's. So this explains why as $\epsilon \rightarrow 0$, we spend almost no time at R .

Intuitively, L is more robust against a variety of opponents' actions than R is, so it's selected by stochastic partial BR learning.

Excellent.